



Fig. 2: Air quality monitors can help keep real-time tab on pollution levels (Credit: dataline.co.ke)

Application areas

As mentioned earlier, IoT solutions help improve the environment at various levels, namely, households, industries and enterprises, or even at city scale.

Industrial and enterprise solutions. Many solutions are available today to help industries and enterprises fight pollution. For instance, Aurassure, an environmental monitoring platform developed by Phoenix Robotix, can be deployed across premises of a factory or office. Solar-powered sensor devices are wall- or pole-mounted across different corners, to create a perimeter around the campus. Sensors pick up environmental data of the surrounding, including air quality (PM 2.5 and PM 10 readings), carbon gas content, other gases, humidity, soil conditions and pH.

Data is stored on a cloud server, which is analysed to create actionable reports. Accessible through a Web-based interface, the operator can log onto the server and study the analytics. In this way, Aurassure helps many industries control emissions and regulate their impact on the environment.

Samantaray says, “Aurassure has been deployed in front of various city municipalities including Rajkot, Siliguri and Coimbatore. In Orissa, we deployed it with industrial customers like Jindal Steel and Dalmia Cements. All of them are

consistently keeping a tab on pollution levels and taking necessary steps after comparing the readings at regular intervals.”

Another example is Wattman platform by Zenatix, which helps geographically-distributed retail and enterprise chains to maintain their appliances and optimise energy consumption. A centralised control system shows real-time electricity usage of all connected appliances—if any appliance is drawing higher energy than it should or is approaching a breakdown, the operator is notified to take preventive measures. This IoT-based building management system aimed towards enterprises has been deployed across various Mother Dairy outlets. It can save at least 30 per cent electricity and considerable carbon emissions.

City-level solutions. Smart cities

are aimed towards environment conservation holistically. Sensorised technologies are deployed across various pollution hotspots like garbage disposal zones, sewage drains, water bodies and busy roads, to actively monitor and mitigate these plans. At city level, a reliable communication network is required, to allow the sensors to communicate with each other.

Mathur explains, “To bring forth a sustainable smart infrastructure to Jamshedpur city, the first step was to bring the whole region under LoRaWAN connectivity. Next, we took different steps to control the various aspects of the environment and created a central visibility platform for all setups from a control office.

“We introduced sensors to sewage drain covers. When sewage becomes almost filled to the brink, the sensors notify the operator at the control to deploy a cleaning team before sewage water overflows. We tested and deployed sensorised bins to notify us in real time when bins across the city get filled, and clean these within time.

“We also introduced sensors to Jamshedpur’s lakes and waterbodies. These read the composition of the water, notify us of any kind of contamination and help us ensure that aquatic flora and fauna are healthy.

“Jamshedpur also utilises smart grids to ensure optimal electricity generation and distribution. Smart meters help us keep a tab on the supply of utilities like electricity and water to reduce wastage.”

In addition to these, smart streetlight control, smart parking and traffic management also factor in a healthy and sustainable urban environment.

Consumer solutions. IoT devices that can help reduce environmental problems from households are aplenty. Smart plugs enable users optimise their usage of electric appliances, especially air-conditioners, geysers and, at times, refrigerators, which generate a high amount of carbon gases and consume a lot of energy.

Smart air purifiers mod-



Fig. 3: Sensor-based water monitoring system in smart cities (Credit: Libelium)